「Def. Let $\beta : I \to \mathbb{R}$ be a unit-speed curve.
A vector field β' is called the unit tangent vector field on β, denote $T = \beta'$.
The derivative $T' = \beta''$ is called the curvature vector field of β.
The value $\kappa(s) = \|T'(s)\|$ $(s \in I)$ is called the curvature function of β.」

To restriction s.t $\kappa > 0$,
the unit-vector field $N = \dfrac{T'}{\kappa}$ is called the principal normal vector field of β.
The cross product $B = T \times N$ on β is called the binormal vector field of β.

「Lemma. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa > 0$,
Then, the three vector fields T, N, and B on β are unit vector fields that are mutually orthogonal at each point.
Proof. By definition, $\|T\| = \|\beta'\| = 1$. And since the curvature $\kappa > 0$, $\|N\| = \frac{1}{\kappa} \cdot \|T'\| = 1$.
And, since $\|T\|^2 = T \cdot T$ is constant, $2T \cdot T' = 0$, so $T \perp N$.
therefore the binormal, $\|B\| = \|T \times N\|^2 = \|T\|^2 \cdot \|N\|^2 - (T \cdot N)^2 = 1 - 0 = 1$. ∎

In summary, $T = \beta'$, $N = T'/\kappa$, $B = T \times N$, and $\{T, N, B\}$ is orthonormal.
Since $B \cdot T = 0$, $B' \cdot T + B \cdot T' = 0$ by differentiate. Therefore,

$$B' \cdot T = -B \cdot T' = -B \cdot \kappa N = 0$$

And since B is unit, $B' \cdot B = 0$, thus, there is a scalar value function τ s.t

$$B' = -\tau N$$

, called the torsion. Now

「Thm. Let β be a unit-speed curve in $\mathbb{R}^3$ with curvature $\kappa > 0$ and torsion τ,

$$\begin{aligned} T' &= \quad\ \kappa N \\ N' &= -\kappa T \qquad + \tau B \\ B' &= \quad -\tau N \end{aligned} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof. The first and third rows are just definition. To show the second, observe
$N' = (N' \cdot T)T + (N' \cdot N)N + (N' \cdot B)B$
Now, $N' \cdot T = N \cdot T'$ and $N' \cdot B = N \cdot B'$ gives the results.